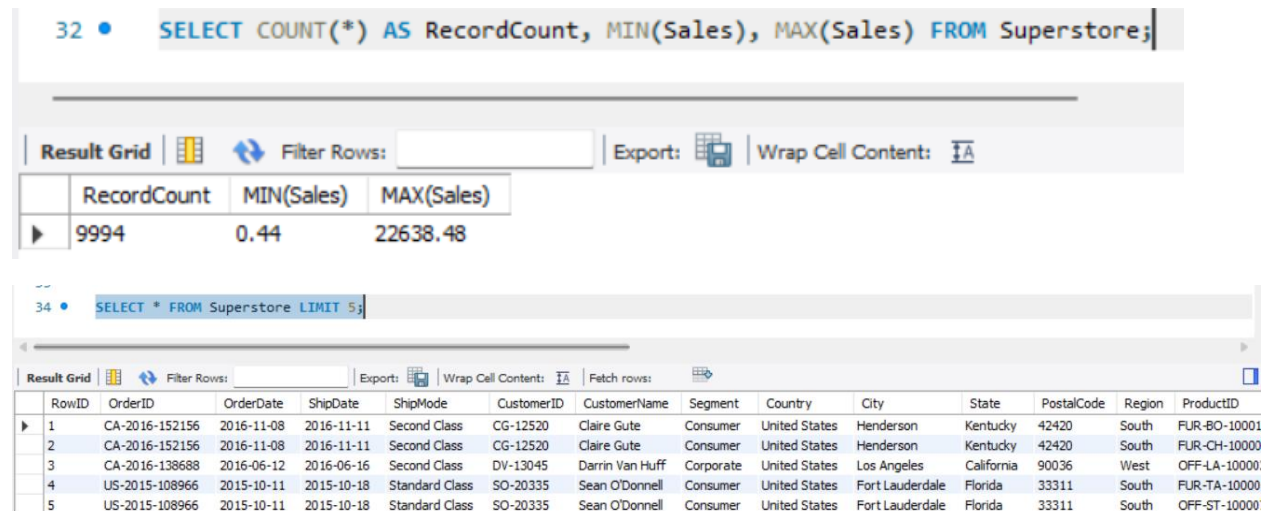


Interactive Data Visualization Dashboard with Risk Monitoring - SQL

STEP 1: SETUP AND DATA IMPORT

- Data Quality Check:**

This gives a quick overview of the dataset, such as the total number of records and the minimum and maximum values in the Sales column. The SELECT * query helps users view sample data and check for any issues with data loading or column mapping.



32 • `SELECT COUNT(*) AS RecordCount, MIN(Sales), MAX(Sales) FROM Superstore;`

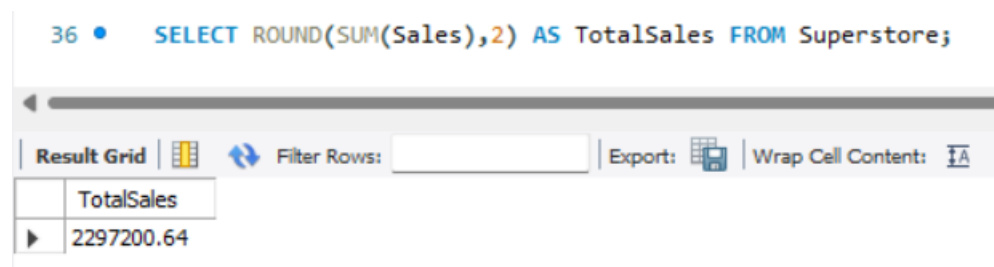
	RecordCount	MIN(Sales)	MAX(Sales)
▶	9994	0.44	22638.48

34 • `SELECT * FROM Superstore LIMIT 5;`

	RowID	OrderID	OrderDate	ShipDate	ShipMode	CustomerID	CustomerName	Segment	Country	City	State	PostalCode	Region	ProductID
▶	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001
	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000
	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000
	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000
	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000

STEP 2: BUSINESS KPIs

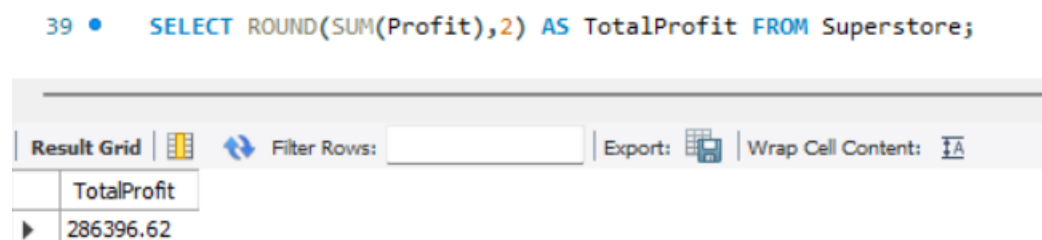
- Total Sales** - This shows the total of the Sales column, rounded to 2 decimal places.



36 • `SELECT ROUND(SUM(Sales),2) AS TotalSales FROM Superstore;`

	TotalSales
▶	2297200.64

- Total Profit** - This shows the total of the Profit column, rounded to 2 decimal places.



39 • `SELECT ROUND(SUM(Profit),2) AS TotalProfit FROM Superstore;`

	TotalProfit
▶	286396.62

- **Profit Margin** – This calculates the profit margin percentage by dividing total profit by total sales and multiplying by 100. The NULLIF function is used to prevent calculation errors when the sales amount is 0.

```
42 • SELECT ROUND((SUM(Profit) / NULLIF(SUM(Sales),0))*100,2) AS ProfitMargin FROM Superstore;
```

ProfitMargin
12.47

- **Order Count** – This counts all the rows in the Superstore table.

```
45 • SELECT COUNT(*) AS OrderCount FROM Superstore;
```

OrderCount
9994

- **Average Order Value** - This divides the total sales by the total number of records in the table to calculate the average order value. If the count is 0, NULLIF helps avoid calculation errors, and the final result is rounded to 2 decimal places.

```
47 • SELECT ROUND(SUM(Sales) / NULLIF(COUNT(*),0),2) AS AvgOrderValue FROM Superstore;
```

AvgOrderValue
229.86

STEP 3: CHART COMPARISON BLOCKS

- **Regional Performance (Bar Chart Equivalent)** - This shows total sales for each region, rounded to 2 decimals, and sorts them from highest to lowest.

```
51 • SELECT Region, ROUND(SUM(Sales),2) AS RegionalSales
52 FROM Superstore
```

Region	RegionalSales
West	725457.83
East	678781.24
Central	501239.74
South	391721.83

- **Category Distribution (Pie Chart Equivalent)** - This shows total sales for each category, grouped by Category and rounded to 2 decimal places.

```

57 • SELECT Category, ROUND(SUM(Sales),2) AS CategorySales
58 FROM Superstore

```

Category	CategorySales
Furniture	741999.73
Office Supplies	719046.85
Technology	836154.06

- **Product Performance Table** - This shows total sales, total profit, and profit margin percentage for each sub-category, grouped by Sub_Category and sorted by total sales in descending order.

```

62 • SELECT Sub_Category, ROUND(SUM(Sales),2) AS TotalSales, ROUND(SUM(Profit),2) AS TotalProfit,
63         ROUND((SUM(Profit)/NULLIF(SUM(Sales),0))*100,2) AS ProfitMargin
64 FROM Superstore
65 GROUP BY Sub_Category
66 ORDER BY TotalSales DESC
67

```

Sub_Category	TotalSales	TotalProfit	ProfitMargin
Phones	330007.10	44515.98	13.49
Chairs	328449.08	26590.12	8.10
Storage	223843.59	21278.95	9.51
Tables	206965.55	-17725.56	-8.56
Binders	203412.64	30221.58	14.86
Machines	189238.64	3384.72	1.79
Accessories	167380.31	41936.72	25.05
Copiers	149528.01	55617.89	37.20
Bookcases	114879.98	-3472.62	-3.02
Appliances	107532.13	18137.97	16.87
Furnishings	91705.12	13059.24	14.24
Paper	78479.24	34053.11	43.39
Supplies	46673.52	-1189.03	-2.55
Art	27118.80	6527.86	24.07
Envelopes	16476.38	6963.99	42.27
Labels	12486.30	5546.18	44.42
Fasteners	3024.25	949.52	31.40

STEP 4: RISK MONITORING

- **Loss-Making Orders** - This counts the total number of orders where profit is less than 0.

```
70 • SELECT COUNT(*) AS LossMakingOrders FROM Superstore WHERE Profit < 0;
```

LossMakingOrders
1871

- **High Discount Orders** - This counts the total number of orders where the discount is greater than 0.6.

```
73 • SELECT COUNT(*) AS HighDiscountOrders FROM Superstore WHERE Discount > 0.6;
```

HighDiscountOrders
718

- **Business Risk Status** - This classifies the business risk as Critical, High, Medium, or Low based on the number of loss-making orders.

```
76 • SELECT
77   CASE
78     WHEN (SELECT COUNT(*) FROM Superstore WHERE Profit < 0) > 500 THEN 'Critical Risk'
79     WHEN (SELECT COUNT(*) FROM Superstore WHERE Profit < 0) > 200 THEN 'High Risk'
80     WHEN (SELECT COUNT(*) FROM Superstore WHERE Profit < 0) > 50 THEN 'Medium Risk'
81     ELSE 'Low Risk'
82   END AS BusinessRiskLevel
```

BusinessRiskLevel
Critical Risk

- **Data Quality Score** - This calculates the total number of orders, identifies risky orders, and then computes a data quality score as a percentage of clean records, rounded to 2 decimal places.

```
85 WITH RiskOrders AS (  
86     SELECT COUNT(*) AS RiskCount  
87     FROM Superstore  
88     WHERE Profit < 0 OR Discount > 0.6  
89 )  
90 SELECT  
91     (SELECT COUNT(*) FROM Superstore) AS TotalOrders,  
92     RiskCount,  
93     ROUND((CAST((SELECT COUNT(*) FROM Superstore) AS FLOAT) - RiskCount) / NULLIF((SELECT COUNT(*) FROM Superstore),0)*100,2)  
94     AS DataQualityScore  
95 FROM RiskOrders;
```

TotalOrders	RiskCount	DataQualityScore
9994	1871	81.28

STEP 5: INTERACTIVE "SLICING" AND SEGMENTATION

- **Segment Slicer** - This shows the number of orders, total sales, and total profit for each segment, grouped by Segment and rounded to 2 decimal places.

```
98 SELECT Segment, COUNT(*) AS Orders, ROUND(SUM(Sales),2) AS TotalSales, ROUND(SUM(Profit),2) AS TotalProfit  
99 FROM Superstore  
100 GROUP BY Segment;
```

Segment	Orders	TotalSales	TotalProfit
Consumer	5191	1161401.14	134118.69
Corporate	3020	706146.33	91979.11
Home Office	1783	429653.17	60298.82

- **Business Risk by Region** - This shows the total orders, number of loss-making orders, and high discount orders for each region, grouped by Region.

```
103 SELECT Region,  
104     COUNT(*) AS Orders,  
105     COUNT(CASE WHEN Profit < 0 THEN 1 END) AS LossOrders,  
106     COUNT(CASE WHEN Discount > 0.6 THEN 1 END) AS HighDiscountOrders  
107 FROM Superstore  
108 GROUP BY Region;
```

Region	Orders	LossOrders	HighDiscountOrders
South	1620	259	132
West	3203	318	100
Central	2323	741	300
East	2848	553	186